

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407296
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Perkins; Richard et al.

Photovoltaic shingles with multi-module power electronics

Abstract

A system includes a plurality of building integrated photovoltaic modules installed on a roof deck and arranged in an array. The plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, with the first building integrated photovoltaic module being vertically adjacent to the second building integrated photovoltaic module. Each of the building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit. The power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells. The power electronics unit is electrically connected to the plurality of solar cells. The power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module.

Inventors: Perkins; Richard (San Jose, CA), Bunea; Gabriela (San Jose, CA), Nguyen; Thierry (San Francisco, CA), Abra; Lewis (San Francisco, CA), Farhangi; Babak (San Jose, CA), Amin; Hasib (Parsippany, NJ), Pham; Henry (San Jose, CA)

Applicant: GAF Energy LLC (Parsippany, NJ)

Family ID: 92460132

Assignee: GAF Energy LLC (Parsippany, NJ)

Appl. No.: 18/949344

Filed: November 15, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250080045 A1	Mar. 06, 2025

Related U.S. Application Data

Publication Classification

Int. Cl.: **H02S40/36** (20140101); **E04D1/30** (20060101); **H02S20/25** (20140101)

U.S. Cl.:

CPC **H02S40/36** (20141201); **E04D1/30** (20130101); **H02S20/25** (20141201); E04D2001/308 (20130101)

Field of Classification Search

CPC: H02S (40/36); H02S (20/25); H02S (20/23); H02S (40/34); E04D (1/30); E04D (2001/308); Y02B (10/10); Y02E (10/50)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1981467	12/1933	Radtke	N/A	N/A
3156497	12/1963	Lessard	N/A	N/A
3581779	12/1970	Gilbert, Jr.	N/A	N/A
4258948	12/1980	Hoffmann	N/A	N/A
4349220	12/1981	Carroll et al.	N/A	N/A
4499702	12/1984	Turner	N/A	N/A
4636577	12/1986	Peterpaul	N/A	N/A
5167579	12/1991	Rotter	N/A	N/A
5437735	12/1994	Younan et al.	N/A	N/A
5590495	12/1996	Bressler et al.	N/A	N/A
5642596	12/1996	Waddington	N/A	N/A
6008450	12/1998	Ohtsuka et al.	N/A	N/A
6033270	12/1999	Stuart	N/A	N/A
6046399	12/1999	Kapner	N/A	N/A
6201180	12/2000	Meyer et al.	N/A	N/A
6220329	12/2000	King et al.	N/A	N/A
6308482	12/2000	Strait	N/A	N/A
6320114	12/2000	Kuechler	N/A	N/A
6320115	12/2000	Kataoka et al.	N/A	N/A
6336304	12/2001	Mimura et al.	N/A	N/A
6341454	12/2001	Koleoglou	N/A	N/A
6407329	12/2001	Iino et al.	N/A	N/A
6576830	12/2002	Nagao et al.	N/A	N/A
6928781	12/2004	Desbois et al.	N/A	N/A
6972367	12/2004	Federspiel et al.	N/A	N/A
7138578	12/2005	Komamine	N/A	N/A
7155870	12/2006	Almy	N/A	N/A

7178295	12/2006	Dinwoodie	N/A	N/A
7487771	12/2008	Eiffert et al.	N/A	N/A
7587864	12/2008	McCaskill et al.	N/A	N/A
7678990	12/2009	McCaskill et al.	N/A	N/A
7678991	12/2009	McCaskill et al.	N/A	N/A
7748191	12/2009	Podirsky	N/A	N/A
7819114	12/2009	Augenbraun et al.	N/A	N/A
7824191	12/2009	Podirsky	N/A	N/A
7832176	12/2009	McCaskill et al.	N/A	N/A
8118109	12/2011	Hacker	N/A	N/A
8168880	12/2011	Jacobs et al.	N/A	N/A
8173889	12/2011	Kalkanoglu et al.	N/A	N/A
8210570	12/2011	Railkar et al.	N/A	N/A
8276329	12/2011	Lenox	N/A	N/A
8312693	12/2011	Cappelli	N/A	N/A
8319093	12/2011	Kalkanoglu et al.	N/A	N/A
8333040	12/2011	Shiao et al.	N/A	N/A
8371076	12/2012	Jones et al.	N/A	N/A
8375653	12/2012	Shiao et al.	N/A	N/A
8404967	12/2012	Kalkanoglu et al.	N/A	N/A
8410349	12/2012	Kalkanoglu et al.	N/A	N/A
8418415	12/2012	Shiao et al.	N/A	N/A
8438796	12/2012	Shiao et al.	N/A	N/A
8468754	12/2012	Railkar et al.	N/A	N/A
8468757	12/2012	Krause et al.	N/A	N/A
8505249	12/2012	Geary	N/A	N/A
8512866	12/2012	Taylor	N/A	N/A
8513517	12/2012	Kalkanoglu et al.	N/A	N/A
8586856	12/2012	Kalkanoglu et al.	N/A	N/A
8601754	12/2012	Jenkins et al.	N/A	N/A
8629578	12/2013	Kurs et al.	N/A	N/A
8646228	12/2013	Jenkins	N/A	N/A
8656657	12/2013	Livsey et al.	N/A	N/A
8671630	12/2013	Lena et al.	N/A	N/A
8677702	12/2013	Jenkins	N/A	N/A
8695289	12/2013	Koch et al.	N/A	N/A
8713858	12/2013	Xie	N/A	N/A
8713860	12/2013	Railkar et al.	N/A	N/A
8733038	12/2013	Kalkanoglu et al.	N/A	N/A
8776455	12/2013	Azoulay	N/A	N/A
8789321	12/2013	Ishida	N/A	N/A
8793940	12/2013	Kalkanoglu et al.	N/A	N/A
8793941	12/2013	Bosler et al.	N/A	N/A
8826607	12/2013	Shiao et al.	N/A	N/A
8835751	12/2013	Kalkanoglu et al.	N/A	N/A
8863451	12/2013	Jenkins et al.	N/A	N/A
8898970	12/2013	Jenkins et al.	N/A	N/A
8925262	12/2014	Railkar et al.	N/A	N/A
8943766	12/2014	Gombarick et al.	N/A	N/A

8946544	12/2014	Jabos et al.	N/A	N/A
8950128	12/2014	Kalkanoglu et al.	N/A	N/A
8959848	12/2014	Jenkins et al.	N/A	N/A
8966838	12/2014	Jenkins	N/A	N/A
8966850	12/2014	Jenkins et al.	N/A	N/A
8994224	12/2014	Mehta et al.	N/A	N/A
9032672	12/2014	Livsey et al.	N/A	N/A
9153950	12/2014	Yamanaka et al.	N/A	N/A
9166087	12/2014	Chihlas et al.	N/A	N/A
9169646	12/2014	Rodrigues et al.	N/A	N/A
9170034	12/2014	Bosler et al.	N/A	N/A
9178465	12/2014	Shiao et al.	N/A	N/A
9202955	12/2014	Livsey et al.	N/A	N/A
9212832	12/2014	Jenkins	N/A	N/A
9217584	12/2014	Kalkanoglu et al.	N/A	N/A
9270221	12/2015	Zhao	N/A	N/A
9273885	12/2015	Rordigues et al.	N/A	N/A
9276141	12/2015	Kalkanoglu et al.	N/A	N/A
9331224	12/2015	Koch et al.	N/A	N/A
9356174	12/2015	Duarte et al.	N/A	N/A
9359014	12/2015	Yang et al.	N/A	N/A
9412890	12/2015	Meyers	N/A	N/A
9528270	12/2015	Jenkins et al.	N/A	N/A
9605432	12/2016	Robbins	N/A	N/A
9711672	12/2016	Wang	N/A	N/A
9755573	12/2016	Livsey et al.	N/A	N/A
9786802	12/2016	Shiao et al.	N/A	N/A
9831818	12/2016	West	N/A	N/A
9912284	12/2017	Svec	N/A	N/A
9923515	12/2017	Rodrigues et al.	N/A	N/A
9938729	12/2017	Coon	N/A	N/A
9991412	12/2017	Gonzalez et al.	N/A	N/A
9998067	12/2017	Kalkanoglu et al.	N/A	N/A
10027273	12/2017	West et al.	N/A	N/A
10115850	12/2017	Rodrigues et al.	N/A	N/A
10128660	12/2017	Apte et al.	N/A	N/A
10156075	12/2017	McDonough	N/A	N/A
10187005	12/2018	Rodrigues et al.	N/A	N/A
10256765	12/2018	Rodrigues et al.	N/A	N/A
10284136	12/2018	Mayfield et al.	N/A	N/A
10454408	12/2018	Livsey et al.	N/A	N/A
10530292	12/2019	Cropper et al.	N/A	N/A
10560048	12/2019	Fisher et al.	N/A	N/A
10563406	12/2019	Kalkanoglu et al.	N/A	N/A
D879031	12/2019	Lance et al.	N/A	N/A
10579028	12/2019	Jacob	N/A	N/A
10693413	12/2019	Rodrigues	N/A	H02S 40/36
10784813	12/2019	Kalkanoglu et al.	N/A	N/A
D904289	12/2019	Lance et al.	N/A	N/A

11012026	12/2020	Kalkanoglu et al.	N/A	N/A
11177639	12/2020	Nguyen et al.	N/A	N/A
11217715	12/2021	Sharenko	N/A	N/A
11251744	12/2021	Bunea et al.	N/A	N/A
11258399	12/2021	Kalkanoglu et al.	N/A	N/A
11283394	12/2021	Perkins et al.	N/A	N/A
11309828	12/2021	Sirski et al.	N/A	N/A
11394344	12/2021	Perkins et al.	N/A	N/A
11424379	12/2021	Sharenko et al.	N/A	N/A
11431280	12/2021	Liu et al.	N/A	N/A
11431281	12/2021	Perkins et al.	N/A	N/A
11444569	12/2021	Clemente et al.	N/A	N/A
11454027	12/2021	Kuiper et al.	N/A	N/A
11459757	12/2021	Nguyen et al.	N/A	N/A
11486144	12/2021	Bunea et al.	N/A	N/A
11489482	12/2021	Peterson et al.	N/A	N/A
11496088	12/2021	Sirski et al.	N/A	N/A
11508861	12/2021	Perkins et al.	N/A	N/A
11512480	12/2021	Achor et al.	N/A	N/A
11527665	12/2021	Boitnott	N/A	N/A
11545927	12/2022	Abra et al.	N/A	N/A
11545928	12/2022	Perkins et al.	N/A	N/A
11658470	12/2022	Nguyen et al.	N/A	N/A
11661745	12/2022	Bunea et al.	N/A	N/A
11689149	12/2022	Clemente et al.	N/A	N/A
11705531	12/2022	Sharenko et al.	N/A	N/A
11728759	12/2022	Nguyen et al.	N/A	N/A
11732490	12/2022	Achor et al.	N/A	N/A
11811361	12/2022	Farhangi et al.	N/A	N/A
11824486	12/2022	Nguyen et al.	N/A	N/A
11824487	12/2022	Nguyen et al.	N/A	N/A
11843067	12/2022	Nguyen et al.	N/A	N/A
2001/0034982	12/2000	Nagao	52/173.3	H02S 20/23
2002/0053360	12/2001	Kinoshita et al.	N/A	N/A
2002/0129849	12/2001	Heckerorth	N/A	N/A
2003/0101662	12/2002	Ullman	N/A	N/A
2003/0132265	12/2002	Villela et al.	N/A	N/A
2003/0217768	12/2002	Guha	N/A	N/A
2004/0000334	12/2003	Ressler	N/A	N/A
2005/0030187	12/2004	Peress et al.	N/A	N/A
2005/0115603	12/2004	Yoshida et al.	N/A	N/A
2005/0144870	12/2004	Dinwoodie	N/A	N/A
2005/0178428	12/2004	Laaly et al.	N/A	N/A
2005/0193673	12/2004	Rodrigues et al.	N/A	N/A
2006/0042683	12/2005	Gangemi	N/A	N/A
2006/0046084	12/2005	Yang et al.	N/A	N/A
2007/0074757	12/2006	Mellott et al.	N/A	N/A
2007/0181174	12/2006	Ressler	N/A	N/A
2007/0193618	12/2006	Bressler et al.	N/A	N/A

2007/0249194	12/2006	Liao	N/A	N/A
2007/0295385	12/2006	Sheats et al.	N/A	N/A
2008/0006323	12/2007	Kalkanoglu et al.	N/A	N/A
2008/0035140	12/2007	Placer et al.	N/A	N/A
2008/0078440	12/2007	Lim et al.	N/A	N/A
2008/0185748	12/2007	Kalkanoglu	N/A	N/A
2008/0271774	12/2007	Kalkanoglu et al.	N/A	N/A
2008/0302030	12/2007	Stancel et al.	N/A	N/A
2008/0315061	12/2007	Fath	N/A	N/A
2009/0000222	12/2008	Kalkanoglu et al.	N/A	N/A
2009/0014057	12/2008	Croft et al.	N/A	N/A
2009/0014058	12/2008	Croft et al.	N/A	N/A
2009/0019795	12/2008	Szacsvey et al.	N/A	N/A
2009/0044850	12/2008	Kimberley	N/A	N/A
2009/0114261	12/2008	Stancel et al.	N/A	N/A
2009/0133340	12/2008	Shiao et al.	N/A	N/A
2009/0159118	12/2008	Kalkanoglu et al.	N/A	N/A
2009/0178350	12/2008	Kalkanoglu et al.	N/A	N/A
2009/0229652	12/2008	Mapel et al.	N/A	N/A
2009/0275247	12/2008	Richter et al.	N/A	N/A
2010/0019580	12/2009	Croft et al.	N/A	N/A
2010/0095618	12/2009	Edison et al.	N/A	N/A
2010/0101634	12/2009	Frank et al.	N/A	N/A
2010/0116325	12/2009	Nikoonahad	N/A	N/A
2010/0131108	12/2009	Meyer	N/A	N/A
2010/0139184	12/2009	Williams et al.	N/A	N/A
2010/0146878	12/2009	Koch et al.	N/A	N/A
2010/0159221	12/2009	Kourtakis et al.	N/A	N/A
2010/0170169	12/2009	Railkar et al.	N/A	N/A
2010/0186798	12/2009	Tormen et al.	N/A	N/A
2010/0242381	12/2009	Jenkins	N/A	N/A
2010/0313499	12/2009	Gangemi	N/A	N/A
2010/0325976	12/2009	DeGenfelder et al.	N/A	N/A
2010/0326488	12/2009	Aue et al.	N/A	N/A
2010/0326501	12/2009	Zhao et al.	N/A	N/A
2011/0030761	12/2010	Kalkanoglu et al.	N/A	N/A
2011/0036386	12/2010	Browder	N/A	N/A
2011/0036389	12/2010	Hardikar et al.	N/A	N/A
2011/0048507	12/2010	Livsey et al.	N/A	N/A
2011/0058337	12/2010	Han et al.	N/A	N/A
2011/0061326	12/2010	Jenkins	N/A	N/A
2011/0100436	12/2010	Cleereman et al.	N/A	N/A
2011/0104488	12/2010	Muessig et al.	N/A	N/A
2011/0132427	12/2010	Kalkanoglu	136/244	H02S 40/34
2011/0168238	12/2010	Metin et al.	N/A	N/A
2011/0239555	12/2010	Cook et al.	N/A	N/A
2011/0302859	12/2010	Crasnianski	N/A	N/A
2011/0314753	12/2010	Farmer et al.	N/A	N/A

2012/0034799	12/2011	Hunt	N/A	N/A
2012/0060434	12/2011	Jacobs	N/A	N/A
2012/0060902	12/2011	Drake	N/A	N/A
2012/0085392	12/2011	Albert et al.	N/A	N/A
2012/0137600	12/2011	Jenkins	N/A	N/A
2012/0176077	12/2011	Oh et al.	N/A	N/A
2012/0212065	12/2011	Cheng et al.	N/A	N/A
2012/0233940	12/2011	Perkins et al.	N/A	N/A
2012/0240490	12/2011	Gangemi	N/A	N/A
2012/0260977	12/2011	Stancel	N/A	N/A
2012/0266942	12/2011	Komatsu et al.	N/A	N/A
2012/0279150	12/2011	Pisllkak et al.	N/A	N/A
2012/0282437	12/2011	Clark et al.	N/A	N/A
2012/0291848	12/2011	Sherman et al.	N/A	N/A
2013/0008499	12/2012	Verger et al.	N/A	N/A
2013/0014455	12/2012	Grieco	N/A	N/A
2013/0118558	12/2012	Sherman	N/A	N/A
2013/0193769	12/2012	Mehta et al.	N/A	N/A
2013/0247988	12/2012	Reese et al.	N/A	N/A
2013/0284267	12/2012	Plug et al.	N/A	N/A
2013/0306137	12/2012	Ko	N/A	N/A
2014/0090697	12/2013	Rodrigues et al.	N/A	N/A
2014/0150843	12/2013	Pearce et al.	N/A	N/A
2014/0173997	12/2013	Jenkins	N/A	N/A
2014/0179220	12/2013	Railkar et al.	N/A	N/A
2014/0182222	12/2013	Kalkanoglu et al.	N/A	N/A
2014/0208675	12/2013	Beerer et al.	N/A	N/A
2014/0254776	12/2013	O'Connor et al.	N/A	N/A
2014/0266289	12/2013	Della Sera et al.	N/A	N/A
2014/0311556	12/2013	Feng et al.	N/A	N/A
2014/0352760	12/2013	Haynes et al.	N/A	N/A
2014/0366464	12/2013	Rodrigues et al.	N/A	N/A
2015/0089895	12/2014	Leitch	N/A	N/A
2015/0162459	12/2014	Lu et al.	N/A	N/A
2015/0340516	12/2014	Kim et al.	N/A	N/A
2015/0349173	12/2014	Morad et al.	N/A	N/A
2016/0105144	12/2015	Haynes et al.	N/A	N/A
2016/0142008	12/2015	Lopez et al.	N/A	N/A
2016/0254776	12/2015	Rodrigues et al.	N/A	N/A
2016/0276508	12/2015	Huang et al.	N/A	N/A
2016/0359451	12/2015	Mao et al.	N/A	N/A
2017/0159292	12/2016	Chihlas et al.	N/A	N/A
2017/0179319	12/2016	Yamashita et al.	N/A	N/A
2017/0179726	12/2016	Garrity et al.	N/A	N/A
2017/0237390	12/2016	Hudson et al.	N/A	N/A
2017/0331415	12/2016	Koppi et al.	N/A	N/A
2018/0094438	12/2017	Wu et al.	N/A	N/A
2018/0097472	12/2017	Anderson et al.	N/A	N/A
2018/0115275	12/2017	Flanigan et al.	N/A	N/A
2018/0254738	12/2017	Yang et al.	N/A	N/A

2018/0294765	12/2017	Friedrich et al.	N/A	N/A
2018/0351502	12/2017	Almy et al.	N/A	N/A
2018/0367089	12/2017	Stutterheim et al.	N/A	N/A
2019/0030867	12/2018	Sun et al.	N/A	N/A
2019/0081436	12/2018	Onodi et al.	N/A	N/A
2019/0123679	12/2018	Rodrigues et al.	N/A	N/A
2019/0253022	12/2018	Hardar et al.	N/A	N/A
2019/0305717	12/2018	Allen et al.	N/A	N/A
2020/0109320	12/2019	Jiang	N/A	N/A
2020/0144958	12/2019	Rodrigues et al.	N/A	N/A
2020/0220819	12/2019	Vu et al.	N/A	N/A
2020/0224419	12/2019	Boss et al.	N/A	N/A
2020/0343397	12/2019	Hem-Jensen	N/A	N/A
2021/0083619	12/2020	Hegedus	N/A	N/A
2021/0115223	12/2020	Bonekamp et al.	N/A	N/A
2021/0159353	12/2020	Li et al.	N/A	N/A
2021/0301536	12/2020	Baggs et al.	N/A	N/A
2021/0343886	12/2020	Sharenko et al.	N/A	N/A
2022/0149213	12/2021	Mensink et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2829440	12/2018	CA	N/A
700095	12/2009	CH	N/A
202797032	12/2012	CN	N/A
217150978	12/2021	CN	N/A
1958248	12/1970	DE	N/A
1039361	12/1999	EP	N/A
1429394	12/2003	EP	H01L 27/142
1837162	12/2006	EP	N/A
1774372	12/2010	EP	N/A
2446481	12/2011	EP	N/A
2533302	12/2011	EP	H01L 31/0201
2784241	12/2013	EP	N/A
3772175	12/2020	EP	N/A
10046767	12/1997	JP	N/A
2002-106151	12/2001	JP	N/A
2001-098703	12/2001	JP	N/A
2017-027735	12/2016	JP	N/A
2018053707	12/2017	JP	N/A
20090084060	12/2008	KR	N/A
10-1348283	12/2013	KR	N/A
10-2019-0000367	12/2018	KR	N/A
10-2253483	12/2020	KR	N/A
2026856	12/2021	NL	N/A
2010/151777	12/2009	WO	N/A
2011/049944	12/2010	WO	N/A
2015/133632	12/2014	WO	N/A

2018/000589	12/2017	WO	N/A
2019/201416	12/2018	WO	N/A
2020-159358	12/2019	WO	N/A
2021-247098	12/2020	WO	N/A

OTHER PUBLICATIONS

Sunflare, Procduts: “Sunflare Develops Prototype for New Residential Solar Shingles”; 2019 <<sunflaresolar.com/news/sunflare-develops-prototype-for-new-residential-solar-shingles>>

retrieved Feb. 2, 2021. cited by applicant

RGS Energy, 3.5KW Powerhouse 3.0 system installed in an afternoon; Jun. 7, 2019

<<facebook.com/RGSEnergy/>> retrieved Feb. 2, 2021. cited by applicant

Tesla, Solar Roof <<tesla.com/solarroof>> retrieved Feb. 2, 2021. cited by applicant

“Types of Roofing Underlayment”, Owens Corning Roofing.

<<<https://www.owenscorning.com/en-us/roofing/tools/how-roofing-underlayment-helps-protect-your-home>>> retrieved. Nov. 1, 2021. cited by applicant

Primary Examiner: Demuren; Babajide A

Attorney, Agent or Firm: GREENBERG TRAURIG, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/584,881, filed Feb. 22, 2024, which is a Section 111(a) application relating to and claiming the benefit of commonly owned, U.S. Provisional Patent Application Ser. No. 63/486,574, filed Feb. 23, 2023, entitled “PHOTOVOLTAIC SHINGLES WITH MULTI-MODULE POWER ELECTRONICS,” the contents of each of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to photovoltaic shingles and, more particularly, photovoltaic shingles with multi-module power electronics.

BACKGROUND

(2) Photovoltaic systems are installed on building roofs to generate electricity.

SUMMARY

(3) In some embodiments, a system includes a plurality of building integrated photovoltaic modules installed on a roof deck, wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck, wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit, wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells, wherein the power electronics unit is electrically connected to the plurality of solar cells, and wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module.

(4) In some embodiments, the power electronics unit is horizontally adjacent to the first end of the one of the at least one row of the plurality of solar cells. In some embodiments, the plurality of

building integrated photovoltaic modules is a plurality of solar shingles. In some embodiments, the at least one row includes a plurality of rows, wherein the plurality of rows includes a first row and a second row above the first row, wherein each of the plurality of building integrated photovoltaic modules includes a first electrical connector proximate to the first end of the first row and electrically connected to the first row of the plurality of solar cells, and a second electrical connector proximate to a first end of the second row and electrically connected to the second row of the plurality of solar cells, and wherein the power electronics unit is connected to the first electrical connector and the second electrical connector.

(5) In some embodiments, the power electronics unit includes a first terminal and a second terminal, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module. In some embodiments, each of the first electrical connector and the second electrical connector includes a first connector and a second connector, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module.

(6) In some embodiments, the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the second electrical connector of the second building integrated photovoltaic module.

(7) In some embodiments, the first terminal of the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module. In some embodiments, the first terminal of the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module. In some embodiments, the second terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the first building integrated photovoltaic module by a first electrical cable. In some embodiments, the second connector of the second electrical connector of the first building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the first building integrated photovoltaic module by a second electrical cable. In some embodiments, the second terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the second building integrated photovoltaic module by a third electrical cable. In some embodiments, the second connector of the second electrical connector of the second building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the second building integrated photovoltaic module by a fourth electrical cable. In some embodiments, the power electronics unit includes a third terminal and a fourth terminal, wherein the third terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the fourth terminal of the power electronics unit of the second building integrated photovoltaic module by a fifth electrical cable.

(8) In some embodiments, the system further includes a third electrical connector installed on the roof deck, wherein each of the power electronics unit of the first building integrated photovoltaic

module and the power electronics unit of the second building integrated photovoltaic module is electrically connected to the third electrical connector. In some embodiments, the third electrical connector includes a first connector and a second connector, wherein the third terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the third electrical connector by a sixth electrical cable. In some embodiments, the fourth terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the third electrical connector by a seventh electrical cable.

(9) In some embodiments, the power electronics unit includes a housing and power electronics within the housing. In some embodiments, the power electronics includes an optimizer, a bypass diode, system monitoring electronic components, a rapid shutdown device, an electronic communication component, or combinations thereof.

(10) In some embodiments, each of the plurality of rows includes a second end opposite the first end of the row, wherein each of the first building integrated photovoltaic module and the second building integrated photovoltaic module includes a first negative electrical terminal located at the first end of the first row thereof, a first positive electrical terminal located at the second end of the second row thereof, a second positive electrical terminal located at the first end of the second row thereof, and a second negative terminal located at the second end of the second row thereof.

(11) In some embodiments, the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners. In some embodiments, the plurality of building integrated photovoltaic modules includes a third building integrated photovoltaic module, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module. In some embodiments, the plurality of building integrated photovoltaic modules includes a fourth building integrated photovoltaic module, wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module. In some embodiments, the fourth building integrated photovoltaic module is vertically adjacent to the third building integrated photovoltaic module. In some embodiments, the power electronics unit of the third building integrated photovoltaic module is electrically connected to the power electronics unit of the fourth building integrated photovoltaic module.

(12) In some embodiments, a system includes a plurality of building integrated photovoltaic modules installed on a roof deck, wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck, wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row; a first power electronics unit, wherein the first power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the first building integrated photovoltaic module, wherein the first power electronics unit is electrically connected to the plurality of solar cells of the first building integrated photovoltaic module; and a second power electronics unit, wherein the second power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the second building integrated photovoltaic module, and wherein the first power electronics unit is electrically connected to the second power electronics unit.

(13) In some embodiments, a method includes obtaining a plurality of building integrated photovoltaic modules, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit, wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells, wherein the power electronics unit is electrically connected to the plurality of solar cells; and installing at least a first building integrated photovoltaic module and a second building

integrated photovoltaic module on a roof deck, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, and wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module.

(14) In some embodiments, the method further includes installing a third building integrated photovoltaic module and a fourth building integrated photovoltaic module on the roof deck, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module, and wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module and vertically adjacent to the third building integrated photovoltaic module. In some embodiments, the plurality of building integrated photovoltaic modules is a plurality of solar shingles. In some embodiments, the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B are a top plan view and a side elevational view, respectively, of some embodiments of a photovoltaic shingle;
- (2) FIG. 2 is a top plan view of some embodiments of a portion of a roofing system including photovoltaic shingles;
- (3) FIG. 3 illustrates some embodiments of a power electronics unit;
- (4) FIG. 4 is a top plan view of some embodiments of a portion of a roofing system including photovoltaic shingles and power electronics units;
- (5) FIGS. 4A through 4C are schematic views of some embodiments of power electronics for photovoltaic shingles;
- (6) FIG. 5 is a top plan view of some embodiments of a photovoltaic module;
- (7) FIG. 6 is a top plan view of some embodiments of a photovoltaic shingle including photovoltaic modules shown in FIG. 5;
- (8) FIG. 7 is a top plan view of some embodiments of a photovoltaic module;
- (9) FIG. 8 is a top plan view of some embodiments of a roofing system including photovoltaic shingles having the photovoltaic modules shown in FIG. 7; and
- (10) FIGS. 9 through 11 are schematic, top plan views of some embodiments of photovoltaic shingles.

DETAILED DESCRIPTION

(11) Referring to FIGS. 1A and 1B, in some embodiments, a photovoltaic shingle **10** includes a first end **12**, a second end **14** opposite the first end **12**, a first edge **13** extending from the first end **12** to the second end **14**, and a second edge **15** opposite the first edge **13** and extending from the first end **12** to the second end **14**. In some embodiments, the photovoltaic shingle **10** includes a headlap portion **16**. In some embodiments, the headlap portion **16** extends from the first end **12** to the second end **14** and from the first edge **13** to a first location **17** between the first edge **13** and the second edge **15**. In some embodiments, the photovoltaic shingle **10** includes a reveal portion **18**. In some embodiments, the reveal portion **18** includes a photovoltaic layer **19**. In some embodiments, the photovoltaic layer **19** includes at least one solar cell **20**. In some embodiments, the photovoltaic shingle **10** includes a first side lap portion **22** located at the first end **12**. In some embodiments, the first side lap portion **22** includes a width extending from the first end **12** to a second location **17a** between the first end **12** and the second end **14**. In some embodiments, the first side lap portion **22** includes a length extending from the first location **17** to the second edge **15**. In some embodiments, the photovoltaic shingle **10** includes a second side lap portion **24** located at the second end **14**. In

some embodiments, the second side lap portion **24** includes a width extending from the second end **14** to a third location **17b** between the first end **12** and the second end **14**. In some embodiments, the second side lap portion **24** includes a length extending from the first location **17** to the second edge **15**. In some embodiments, the photovoltaic shingle **10** includes an outer surface **25** and an inner surface **27** opposite the outer surface **25**. In some embodiments, the reveal portion **18** extends from the first side lap portion **22** to the second side lap portion **24** and from the second edge **15** to the first location **17**. In some embodiments, the photovoltaic shingle **10** is configured to be installed on a building structure. In some embodiments, the photovoltaic shingle **10** is configured to be installed on a roof deck. In some embodiments, at least one junction box **26** is located on the first side lap portion **22**. In some embodiments, the at least one junction box **26** includes a plurality of the junction boxes **26**. In certain embodiments, other electronic and electrical components may be attached to the first side lap portion **22**. In some embodiments, non-limiting examples of such electronic and electrical components include an electrical connector, a rapid shutdown device, an optimizer, and an inverter. In some embodiments, the photovoltaic shingle **10** includes a structure, composition, components, and/or function similar to those of one or more embodiments of the photovoltaic shingles disclosed in in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled “Building Integrated Photovoltaic System,” owned by GAF Energy LLC, and U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled “Photovoltaic Shingles and Methods of Installing Same,” the contents of each of which are incorporated by reference herein in their entirety.

(12) In some embodiments, the at least one solar cell **20** includes a plurality of the solar cells **20**. In some embodiments, the plurality of solar cells **20** includes two solar cells. In some embodiments, the plurality of solar cells **20** includes three solar cells. In some embodiments, the plurality of solar cells **20** includes four solar cells. In some embodiments, the plurality of solar cells **20** includes five solar cells. In some embodiments, the plurality of solar cells **20** includes six solar cells. In some embodiments, the plurality of solar cells **20** includes seven solar cells. In some embodiments, the plurality of solar cells **20** includes eight solar cells. In some embodiments, the plurality of solar cells **20** includes nine solar cells. In some embodiments, the plurality of solar cells **20** includes ten solar cells. In some embodiments, the plurality of solar cells **20** includes eleven solar cells. In some embodiments, the plurality of solar cells **20** includes twelve solar cells.

(13) In some embodiments, the plurality of solar cells **20** includes thirteen solar cells. In some embodiments, the plurality of solar cells **20** includes fourteen solar cells. In some embodiments, the plurality of solar cells **20** includes fifteen solar cells. In some embodiments, the plurality of solar cells **20** includes sixteen solar cells. In some embodiments, the plurality of solar cells **20** includes more than sixteen solar cells.

(14) In some embodiments, the plurality of solar cells **20** is arranged in one row (i.e., one reveal). In another embodiment, the plurality of solar cells **20** is arranged in two rows (i.e., two reveals). In another embodiment, the plurality of solar cells **20** is arranged in three rows (i.e., three reveals). In another embodiment, the plurality of solar cells **20** is arranged in four rows (i.e., four reveals). In another embodiment, the plurality of solar cells **20** is arranged in five rows (i.e., five reveals). In another embodiment, the plurality of solar cells **20** is arranged in six rows (i.e., six reveals). In other embodiments, the plurality of solar cells **20** is arranged in more than six rows. In some embodiments, each of the rows of the plurality of solar cells **20** includes a first end **28** located proximate to the first side lap portion **22**.

(15) In some embodiments, a roofing system includes a plurality of the photovoltaic shingles **10** installed on a roof deck. In some embodiments, the roof deck is a steep slope roof deck. As defined herein, a “steep slope roof deck” is any roof deck that is disposed on a roof having a pitch of Y/X, where Y and X are in a ratio of 4:12 to 12:18, where Y corresponds to the “rise” of the roof, and where X corresponds to the “run” of the roof.

(16) In some embodiments, the plurality of photovoltaic shingles **10** is installed directly to the roof

deck. In some embodiments, each of the plurality of photovoltaic shingles **10** is installed on the roof deck by a plurality of fasteners. In some embodiments, the plurality of fasteners is installed through the headlap portion **16**. In some embodiments, the plurality of fasteners includes a plurality of nails. In some embodiments, the plurality of fasteners includes a plurality of rivets. In some embodiments, the plurality of fasteners includes a plurality of screws. In some embodiments, the plurality of fasteners includes a plurality of staples.

(17) In some embodiments, each of the plurality of photovoltaic shingles **10** is installed on the roof deck by an adhesive. In some embodiments, the adhesive is adhered directly to the roof deck. In some embodiments, the adhesive is adhered to an underlayment. In some embodiments, the underlayment is adhered directly to the roof deck. In some embodiments, the adhesive is located on a rear surface of the photovoltaic shingle **10**. In some embodiments, the adhesive includes at least one adhesive strip. In some embodiments, the adhesive includes a plurality of adhesive strips. In some embodiments, the plurality of adhesive strips is arranged intermittently. In some embodiments, the adhesive is located proximate to one edge of the photovoltaic shingle **10**. In some embodiments, the adhesive is a peel and stick film sheet. In some embodiments, the peel and stick film sheet includes at least one sheet of film removably attached to the rear surface. In some embodiments, the peel and stick film sheet is composed of EverGuard Freedom HW peel and stick membrane manufactured by GAF. In some embodiments, the adhesive includes polyvinyl butyrate, acrylic, silicone, or polycarbonate. In some embodiments, the adhesive includes pressure sensitive adhesives.

(18) In some embodiments, a first photovoltaic shingle **10** is vertically adjacent to a second photovoltaic shingle **10**. In some embodiments, the first photovoltaic shingle **10** overlays at least a part of the headlap portion **16** of the second photovoltaic shingle **10**.

(19) Referring to FIG. 2, in some embodiments, a roofing system includes a plurality of photovoltaic shingles **10** installed on a roof deck **75** in a plurality of rows **R**. In some embodiments, each of a plurality of the photovoltaic shingles **10** includes a first electrical connector **30**. In some embodiments, the first electrical connector **30** is located on the first side lap portion **22**. In some embodiments, the first electrical connector **30** is located on the first side lap portion **22** proximate to the first end **28** of the row of solar cells **20**. In some embodiments, the first electrical connector **30** is at least partially embedded within the side lap portion **22**. In some embodiments, the first electrical connector **30** is attached to an upper surface of the side lap portion **22**. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by an adhesive. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by welding. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by heat welding. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by ultrasonic welding. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by at least one fastener. In some embodiments, the first electrical connector **30** is removably attached to the upper surface of the side lap portion **22**.

(20) In some embodiments, the first electrical connector **30** includes a structure, composition, components, and/or function similar to those of one or more embodiments of the electrical connectors disclosed in U.S. Patent Application Publication No. 2022/0029037 to Nguyen et al., published Jan. 27, 2022 and entitled “Photovoltaic Systems” and U.S. Patent Application Publication No. 2022/0311377 to Bunea et al., published Sep. 29, 2022 and entitled “Electrical Components for Photovoltaic Systems,” the contents of each of which are incorporated by reference herein in its entirety.

(21) In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 7 mm. In some

(23) In some embodiments, the first electrical connector **30** is electrically connected to the plurality of solar cells **20**. In some embodiments, the first electrical connector **30** is electrically connected to the plurality of solar cells **20** by electrical bussing. In some embodiments, the electrical bussing is embedded within the photovoltaic shingle **10**.

(24) In some embodiments, the first electrical connector **30** includes a housing **32** and a pair of connectors **34a**, **34b**. In some embodiments, the connectors **34a**, **34b** extend from the housing **32**. In some embodiments, the connectors **34a**, **34b** are snap-fit connectors. In some embodiments, the connector **34a** is spaced apart from the connector **34b**. In some embodiments, the connector **34a** is located proximate to one end of the housing **32** and the connector **34b** is located proximate to an opposite end of the housing **32**. In some embodiments, each of the first electrical connectors **30** is electrically connected to one another by a plurality of electrical jumper wires **36** (also referred to as jumper wires, electrical jumper cables, or jumper cables). In some embodiments, the first electrical connectors **30** are electrically connected to one another in series by the plurality of jumper wires **36**. In some embodiments, each of the jumper wires **36** includes a first connector **38a** at a first end thereof and a second connector **38b** at a second, opposite end thereof. In some embodiments, the first connector **38a** of a corresponding one of the jumper wires **36** is connected to a corresponding one of the connector **34b** of the housing **32** of the first electrical connector **30**. In some embodiments, the first connector **38a** of a corresponding one of the jumper wires **36** is preconnected to a corresponding one of the connector **34b** of the housing **32** of the first electrical connector **30** of the photovoltaic shingle **10** prior to installation on the roof deck **75**. In some embodiments, the second connector **38b** of a corresponding one of the jumper wires **36** is connected to a corresponding one of the connector **34a** of the housing **32** of the first electrical connector **30** of an adjacent lower one of the plurality of photovoltaic shingles **10** in rows **R2**, **R3**, **R4**.

(25) In some embodiments, the system includes a second electrical connector **40**. In some embodiments, the second electrical connector **40** is installed on the roof deck **75**. In some embodiments, the second electrical connector **40** is located below row **R4**. In some embodiments, the second electrical connector **40** is proximate to row **R4**. In some embodiments, the second electrical connector **40** is adjacent row **R4**. In some embodiments, the second electrical connector **40** includes a housing **42** and a pair of connectors **44a**, **44b**. In some embodiments, the connectors **44a**, **44b** extend from the housing **42**. In some embodiments, the second electrical connector **40** has a similar structure and/or function of those of the first electrical connector **30**. In some embodiments, a return wire **46** electrically connects the first electrical connector **30** of the photovoltaic shingle **10** in the uppermost row **R1** with the second electrical connector **40**. In some embodiments, the return wire **46** includes a first connector **48a** at a first end thereof and a second connector **48b** at a second, opposite end thereof. In some embodiments, the connector **48a** of the return wire **46** is connected to the connector **34a** of the first electrical connector **30** of the photovoltaic shingle **10** in row **R1**. In some embodiments, the connector **48b** of the return wire **46** is connected to the connector **44b** of the second electrical connector **40**. In some embodiments, the return wire **46** has a length. In some embodiments, the length of the return wire **46** may vary and be selected by an installer of the system, depending upon the number of rows **R** of the photovoltaic shingles **10** in the column of the array thereof.

(26) Referring to FIG. 3, in some embodiments, the photovoltaic shingle **10** includes a power electronics unit **50**. In some embodiments, the power electronics unit **50** is located on the first side lap portion **22**. In some embodiments, the power electronics unit **50** is attached to the first side lap portion **22**. In some embodiments, the power electronics unit **50** is laminated with the first side lap portion **22**. In some embodiments, the power electronics unit **50** is a DC optimizer PCB. In some embodiments, the power electronics unit **50** may be other PCB's having other functionality, such as bypass diodes, system monitoring electronic components, electronic communication components, rapid shutdown devices, or electronic circuitry assemblies performing one or more functionalities

such as power optimization, rapid-shutdown, monitoring or communication.

(27) Referring to FIG. 4, in some embodiments, the roofing system illustrated in FIG. 3 and described above may be configured to incorporate a plurality of the power electronics units 50. In some embodiments, the power electronics units 50 are installed once every two rows of the photovoltaic shingles 10. In some embodiments, the power electronics units 50 are installed in every row of the photovoltaic shingles 10. In some embodiments, the power electronics units 50 are installed in some of the rows of the photovoltaic shingles 10. In some embodiments, the power electronics unit 50 includes a housing 52. In some embodiments, the housing 52 is composed of plastic. In some embodiments, the housing is composed of metal. In some embodiments, power electronics are sealed within the housing 52. In some embodiments, the power electronics unit 50 includes four terminals 54a, 54b, 54c, 54d. In some embodiments, each of the terminals 54a-d is a bulkhead terminal. In some embodiments, the power electronics unit 50 may include less than four of the terminals 54a-d. In some embodiments, the power electronics unit 50 may include more than four of the terminals 54a-d.

(28) In some embodiments, a first power electronics unit 50a is installed on the photovoltaic shingle 10 in row R2. In some embodiments, the first power electronics unit 50a is connected to the first electrical connector 30 of the photovoltaic shingle 10 in row R2. In some embodiments, the terminal 54a of the first power electronics unit 50a is connected to connector 34a of the first electrical connector 30 of the photovoltaic shingle 10 in row R2. In some embodiments, the second connector 38b of jumper wire 36a is connected to the terminal 54b of the first power electronics unit 50a, while the first connector 38a of the jumper wire 36a is connected to the connector 34a of the first electrical connector 30 of the photovoltaic shingle 10 in row R1. In some embodiments, the second connector 38b of jumper wire 36b is connected to the first connector 34a of the connector 30 in row R2, while the first connector 38a of the jumper wire 36b is connected to the second connector 34b of the connector 30 in row R1.

(29) In some embodiments, a second power electronics unit 50b is installed on the photovoltaic shingle 10 in row R4. In some embodiments, the second power electronics units 50b is connected to the first electrical connector 30 of the photovoltaic shingle 10 in row R4. In some embodiments, the terminal 54a of the second power electronics unit 50b is connected to connector 34a of the first electrical connector 30 of the photovoltaic shingle 10 in row R4. In some embodiments, the second connector 38b of jumper wire 36c is connected to the terminal 54b of the second power electronics unit 50b, while the first connector 38a of the jumper wire 36c is connected to the connector 34a of the first electrical connector 30 of the photovoltaic shingle 10 in row R3. In some embodiments, the second connector 38b of jumper wire 36d is connected to the first connector 34a of the connector 30 in row R4, while the first connector 38a of the jumper wire 36d is connected to the second connector 34b of the connector 30 in row R3.

(30) In some embodiments, the terminal 54c of the first power electronics unit 50a is electrically connected to terminal 54d of the second power electronics unit 50b. In some embodiments, the terminal 54c of the first power electronics unit 50a is electrically connected to terminal 54d of the second power electronics unit 50b by a jumper wire 56. In some embodiments, the terminal 54d of the first power electronics unit 50a is electrically connected to the connector 44b of the second electrical connector 40. In some embodiments, the terminal 54d of the first power electronics unit 50a is electrically connected to the connector 44b of the second electrical connector 40 by a jumper wire 58. In some embodiments, the terminal 54c of the second power electronics unit 50b is electrically connected to the connector 44a of the second electrical connector 40. In some embodiments, the terminal 54c of the second power electronics unit 50b is electrically connected to the connector 44a of the second electrical connector 40 by a jumper wire 60. In some embodiments, the first power electronics unit 50a is electrically connected to the second power electronics unit 50b. In some embodiments, it should be understood that the roofing system may have more or less than four of the rows R1-R4 of the photovoltaic shingles 10, and, therefore, the

configuration of the power electronics units **50a-b**, the electrical connectors **30**, **40** and wires **36a-d**, **56**, **58**, **60** would be modified accordingly.

(31) In some embodiments, any or all of the power electronics units **50a-b**, the electrical connectors **30**, **40** and wires **36a-d**, **56**, **58**, **60** may be installed by an installer on the roof deck **75**. In some embodiments, any or all of the power electronics units **50a-b**, the electrical connectors **30**, **40** and wires **36a-d**, **56**, **58**, **60** may be prefabricated prior to installation by an installer on the roof deck **75**. In some embodiments, at least two of the photovoltaic shingles **10** are mechanically attached to one another and to a common headlap portion using adhesives, bonding or thermal welding processes. In some embodiments, the power electronics units **50a-b** and applicable ones of the wires **36a-d**, **56**, **58**, **60** are attached to the photovoltaic shingles **10** that are on the common headlap portion.

(32) Referring to FIG. **4A**, in some embodiments, each of the power electronics units **50** is located within a corresponding one of the junction boxes **26**. In some embodiments, the terminals **54a**, **54b** are located on or in the junction box **26**. In some embodiments, at least one of the power electronics units **50** is located within a corresponding one of the junction boxes **26**. Referring to FIG. **4B**, in some embodiments, each of the power electronics units **50** is separate from the terminals **54a**, **54b**. Referring to FIG. **4C**, in some embodiments, the power electronics unit **50** includes one or more electrical connections **55** to the photovoltaic module **10**, the solar cells **20**, circuit formation, or other circuitry in between.

(33) Referring to FIG. **5**, in some embodiments, a photovoltaic module **110** includes a first end **112**, a second end **114** opposite the first end **112**, a reveal portion **118** having a plurality of solar cells **120**, a first side lap portion **122** at the first end **112**, and a second side lap portion **124** at the second end **114**. In some embodiments, the photovoltaic module **110** includes a busbar **125** extending from the first side lap portion **122** to the second side lap portion **124**. In some embodiments, the busbar **125** includes a first branch **127** located at the first side lap portion **122**. In some embodiments, the busbar **125** includes a second branch **129** located at the second side lap portion **124**. In some embodiments, the first branch **127** includes a negative electrical terminal. In some embodiments, the second branch **129** includes a positive electrical terminal. In some embodiments, the first branch **127** includes a positive electrical terminal. In some embodiments, the second branch **129** includes a negative electrical terminal. In some embodiments, the photovoltaic module **110** includes at least one bypass diode **130** electrically connected to the busbar **125**. In some embodiments, the at least one bypass diode **130** includes a plurality of bypass diodes **130**. In some embodiments, the at least one bypass diode **130** is located within a section **131** of the photovoltaic module **110** that is located between a first edge **113** of the photovoltaic module **110** and the plurality of solar cells **120**.

(34) Referring to FIG. **6**, in some embodiments, a photovoltaic shingle **100** includes a plurality of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** is attached to a backsheet **134**. In some embodiments, the plurality of photovoltaic modules **110** overlay the backsheet **134**. In some embodiments each of the plurality of photovoltaic modules **110** is vertically stacked relative to one another on the backsheet **134**. In some embodiments, the photovoltaic shingle **100** includes a headlap portion **116**. In some embodiments, a portion of the backsheet **134** forms the headlap portion **116**. In some embodiments, the backsheet **134** includes at least one layer. In some embodiments, the backsheet **134** includes at least two layers. In some embodiments, the backsheet **134** is composed of a polymeric material. In some embodiments, In some embodiments, the backsheet **134** is composed of polyethylene terephthalate (“PET”). In some embodiments, the backsheet **134** is composed of ethylene tetrafluoroethylene (“ETFE”). In some embodiments, the backsheet **134** is composed of an acrylic such as polymethyl methacrylate (“PMMA”). In some embodiments, the backsheet **134** is composed of thermoplastic polyolefin (TPO). In some embodiments, the backsheet **134** is composed of a single ply TPO roofing membrane. In some embodiments, non-limiting examples of TPO membranes are disclosed in U.S.

Pat. No. 9,359,014 to Yang et al., which is incorporated by reference herein in its entirety. In some embodiments, the backsheet **134** is composed of polyvinyl chloride. In some embodiments, the backsheet **134** is composed of ethylene propylene diene monomer (EPDM) rubber. In some embodiments, the backsheet **134** includes a flame retardant additive. In some embodiments, the flame retardant additive may be clays, nanoclays, silicas, carbon black, metal hydroxides such as aluminum hydroxide, metal foils, graphite, and combinations thereof.

(35) In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134**. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by an adhesive. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by thermal bonding. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by ultrasonic welding. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by thermal welding. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by mechanical fasteners. In some embodiments, the fasteners may include nails, screws, rivets, or staples.

(36) Still referring to FIG. 6, in some embodiments, a first photovoltaic module **110a** of the photovoltaic modules **110** is oriented such that it includes a negative terminal at the first end **112** and a positive terminal at the second end **114**. In some embodiments, a second photovoltaic module **110b** of the photovoltaic modules **110** is oriented 180 degrees relative to the first photovoltaic module **110a** such that includes a positive terminal at the second end **114** (now on the “left” side as shown in FIG. 6) and a negative terminal at the first end **112** (now on the “right” side as shown in FIG. 6). In some embodiments, a power electronics unit **150** is electrically connected to the first branch **127** (negative terminal) of the first photovoltaic module **110a** and the second branch **129** (positive terminal) of the second photovoltaic module **110b**. In some embodiments, the power electronics unit **150** is electrically connected to the first and second photovoltaic modules **110a**, **110b** in a manner similar to that as described above with respect to the power electronics units **50** to the photovoltaic shingles **10**. In some embodiments, the second branch **129** (positive terminal) of the first photovoltaic module **110a** is electrically connected to the first branch **127** (negative terminal) of the second photovoltaic module **110b**. In some embodiments, the second branch **129** (positive terminal) of the first photovoltaic module **110a** is electrically connected to the first branch **127** (negative terminal) of the second photovoltaic module **110b** by a jumper wire **136** (electrical jumper cable).

(37) In some embodiments, a lower edge **115** of the photovoltaic module **110b** overlays the section **131** of the first photovoltaic module **110a**. In some embodiments, the lower edge **115** of the photovoltaic module **110b** overlays the at least one bypass diode **130** of the first photovoltaic module **110a** to reduce wasted space resulting from the section **131**.

(38) In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes two of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes three of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes four of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes five of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes six of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes more than six of the photovoltaic modules **110**.

(39) FIG. 7 illustrates the photovoltaic shingle **100** having the photovoltaic module **110b** with the string of the solar cells **120** thereof reversed. In some embodiments, the string of solar cells **120** is reversed during lamination of the photovoltaic module **110b**. In some embodiments, the first photovoltaic module **110a** overlaps the section **131** containing the at least one bypass diode **130**. In some embodiments, the first photovoltaic module **110a** overlaps the section **131** thereof containing

the at least one bypass diode **130** to reduce wasted space. FIG. **8** illustrates a plurality of the photovoltaic modules **110** installed on the roof deck with second photovoltaic module **110b** having the reversed strings of solar cells **120**.

(40) Referring to FIG. **9**, in some embodiments, a photovoltaic shingle **200** includes at least two photovoltaic modules **210a**, **210b** attached to backsheet **234**. In some embodiments, the backsheet **234** forms a headlap portion **216**. In some embodiments, each of the photovoltaic modules **210a**, **210b** includes a first end **212**, a second end **214** opposite the first end **212**, a first positive terminal **220a** and a first negative terminal **220b** proximate the first end **212**, and a second positive terminal **222a** and a second negative terminal **222b** proximate the second end **214**. In some embodiments, the second photovoltaic module **210b** is vertically stacked and oriented 180 degrees relative to the first photovoltaic module **210a**. Referring to FIG. **10**, in some embodiments, the photovoltaic shingle **200** includes a third photovoltaic module **210c** attached to a backsheet **234**. In some embodiments, the third photovoltaic module **210c** is vertically stacked and in the same orientation as the first photovoltaic module **210a**. Referring to FIG. **11**, in some embodiments, the second photovoltaic module **210b** is horizontally positioned adjacent to, and oriented 180 degrees relative to, the first photovoltaic module **210a**.

Claims

1. A system, comprising: a plurality of building integrated photovoltaic modules installed on a roof deck, wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck, wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit, wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells, wherein the power electronics unit is electrically connected to the plurality of solar cells, wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module, wherein the power electronics unit includes a first terminal and a second terminal, a first electrical connector electrically connected to the plurality of solar cells, and a second electrical connector electrically connected to the plurality of solar cells, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module.
2. The system of claim 1, wherein the power electronics unit is horizontally adjacent to the first end of the one of the at least one row of the plurality of solar cells.
3. The system of claim 1, wherein the plurality of building integrated photovoltaic modules is a plurality of solar shingles.
4. The system of claim 1, wherein each of the first electrical connector and the second electrical connector includes a first connector and a second connector, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module.
5. The system of claim 4, wherein the power electronics unit of the first building integrated

photovoltaic module is mechanically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the second electrical connector of the second building integrated photovoltaic module.

6. The system of claim 5, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module.

7. The system of claim 6, wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module.

8. The system of claim 7, wherein the second terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the first building integrated photovoltaic module by a first electrical cable.

9. The system of claim 8, wherein the second connector of the second electrical connector of the first building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the first building integrated photovoltaic module by a second electrical cable.

10. The system of claim 9, wherein the second terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the second building integrated photovoltaic module by a third electrical cable.

11. The system of claim 10, wherein the second connector of the second electrical connector of the second building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the second building integrated photovoltaic module by a fourth electrical cable.

12. The system of claim 11, wherein the power electronics unit includes a third terminal and a fourth terminal, wherein the third terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the fourth terminal of the power electronics unit of the second building integrated photovoltaic module by a fifth electrical cable.

13. The system of claim 12, further comprising a third electrical connector installed on the roof deck, wherein each of the power electronics unit of the first building integrated photovoltaic module and the power electronics unit of the second building integrated photovoltaic module is electrically connected to the third electrical connector.

14. The system of claim 13, wherein the third electrical connector includes a first connector and a second connector, wherein the third terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the third electrical connector by a sixth electrical cable.

15. The system of claim 14, wherein the fourth terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the third electrical connector by a seventh electrical cable.

16. The system of claim 1, wherein the power electronics unit includes a housing and power electronics within the housing.

17. The system of claim 16, wherein the power electronics includes an optimizer, a bypass diode, system monitoring electronic components, a rapid shutdown device, an electronic communication component, or combinations thereof.

18. The system of claim 1, wherein the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners.

19. The system of claim 1, wherein the plurality of building integrated photovoltaic modules includes a third building integrated photovoltaic module, wherein the third building integrated

photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module.

20. The system of claim 19, wherein the plurality of building integrated photovoltaic modules includes a fourth building integrated photovoltaic module, wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module.

21. The system of claim 20, wherein the fourth building integrated photovoltaic module is vertically adjacent to the third building integrated photovoltaic module.

22. The system of claim 21, wherein the power electronics unit of the third building integrated photovoltaic module is electrically connected to the power electronics unit of the fourth building integrated photovoltaic module.

23. A system, comprising: a plurality of building integrated photovoltaic modules installed on a roof deck, wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck, wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, a first electrical connector electrically connected to the plurality of solar cells, and a second electrical connector electrically connected to the plurality of solar cells; a first power electronics unit, wherein the first power electronics unit includes a first terminal and a second terminal, wherein the first power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the first building integrated photovoltaic module, wherein the first power electronics unit is electrically connected to the plurality of solar cells of the first building integrated photovoltaic module; and a second power electronics unit, wherein the second power electronics unit includes a first terminal and a second terminal, wherein the second power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the second building integrated photovoltaic module, and wherein the first power electronics unit is electrically connected to the second power electronics unit, and wherein the first terminal of the first power electronics unit is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module.

24. A method, comprising: obtaining a plurality of building integrated photovoltaic modules, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit, wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells, wherein the power electronics unit is electrically connected to the plurality of solar cells wherein the power electronics unit includes a first terminal and a second terminal, a first electrical connector electrically connected to the plurality of solar cells, and a second electrical connector electrically connected to the plurality of solar cells; and installing at least a first building integrated photovoltaic module and a second building integrated photovoltaic module on a roof deck, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, and wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module.

25. The method of claim 24, further comprising installing a third building integrated photovoltaic module and a fourth building integrated photovoltaic module on the roof deck, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module, and wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module and vertically adjacent to the third building integrated photovoltaic module.
26. The method of claim 24, wherein the plurality of building integrated photovoltaic modules is a plurality of solar shingles.
27. The method of claim 24, wherein the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners.
28. The system of claim 1, wherein the at least one row includes a plurality of rows, wherein the plurality of rows includes a first row and a second row above the first row, wherein the first electrical connector is proximate to the first end of the first row and electrically connected to the first row of the plurality of solar cells, and the second electrical connector is proximate to a first end of the second row and electrically connected to the second row of the plurality of solar cells.
29. The system of claim 28, wherein each of the plurality of rows includes a second end opposite the first end of the row, wherein each of the first building integrated photovoltaic module and the second building integrated photovoltaic module includes a first negative electrical terminal located at the first end of the first row thereof, a first positive electrical terminal located at the second end of the second row thereof, a second positive electrical terminal located at the first end of the second row thereof, and a second negative terminal located at the second end of the second row thereof.
30. The system of claim 23, wherein each of the first power electronics unit and second power electronics unit includes an optimizer, a bypass diode, system monitoring electronic components, a rapid shutdown device, an electronic communication component, or combinations thereof.
-